



HIGH PERFORMANCE COMPUTING • TACC

Submitting, Monitoring & Canceling Jobs on TACC

Running batch jobs with SLURM · sbatch, squeue & scancel · plus the TACC Analysis Portal

What we'll cover

1

SUBMIT

Stage your files, write an SBATCH script, and hand it to the scheduler with sbatch.

2

MONITOR

Watch your job in the queue with squeue and showq, and read its status.

3

CANCEL

Pull a queued or running job back with scancel — by ID, name, or user.

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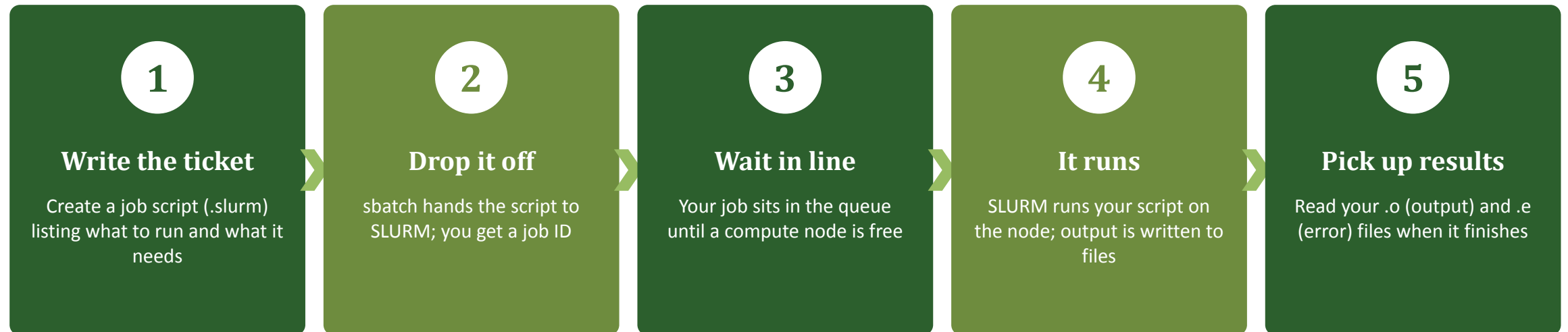
TAP

Skip the scripts entirely: launch Jupyter, RStudio, or a desktop from the browser.

How a batch job works

You don't run heavy work directly on the login node. Instead you write down instructions, hand them to a scheduler (SLURM), and it runs your job on a compute node when resources free up.

Think of it like dropping clothes at a dry cleaner:



First: run your work on \$SCRATCH

\$HOME and \$WORK are for storing and keeping track of important items. The actual job activity — reading and writing to disk — should be offloaded to your resource's \$SCRATCH file system.

\$HOME

Small, backed-up space for config and source. Not for heavy job I/O.

\$WORK

Larger storage for datasets and originals you want to keep.

\$SCRATCH

Fast, large space where jobs should actually read and write. Not backed up.

Rule of thumb: start a job from anywhere, but do the work on \$SCRATCH. Keep originals on \$HOME or \$WORK so you can copy them back over if you need to regenerate results.

1

Submit

Stage your files, describe the job in an SBATCH script, and send it to SLURM.

1 Step 1 — Copy your files into TACC

If your code or data isn't on TACC yet, copy it up with scp (secure copy) from your own machine's terminal:

```
scp -r local/folder your-username@vista.tacc.utexas.edu:
```

Reading the command:

- -r copies a whole folder (drop it for a single file).
- The trailing colon (:) means “land this in my home directory on Vista.” Add a path after it to place it elsewhere.
- Swap vista.tacc.utexas.edu for whichever system you're using (frontera, ls6, stampede3, ...).
- Windows users without scp can use Git Bash or an SFTP client such as WinSCP.

2 Step 2 — Write the SBATCH script

A job script is a normal shell script. The #SBATCH lines at the top are directives that tell SLURM what your job needs; the commands below them are what actually runs.

```
#!/bin/bash
#SBATCH -J myjob           # Job name
#SBATCH -o myjob.o%j      # stdout file (%j = job ID)
#SBATCH -e myjob.e%j      # stderr / error file
#SBATCH -p small          # Queue (partition) name
#SBATCH -N 1              # Total nodes (1 for serial)
#SBATCH -n 1              # Total MPI tasks (1 for serial)
#SBATCH -t 01:30:00       # Run time (hh:mm:ss)
#SBATCH -A myproject      # Allocation (if you have >1)
#SBATCH --mail-type=all   # Email at begin & end
#SBATCH --mail-user=you@utexas.edu # Where to send it

module load python3
python3 my_code.py
```

3 Step 3 — Submit it with sbatch

Hand the script to SLURM. It returns a job ID you'll use to monitor or cancel the job.

```
$ sbatch myjob.slurm  
Submitted batch job 1234567
```

Write down that number

1234567 is your job ID — you'll need it for squeue and scancel.

Heads up: you cannot increase a job's run time (-t) once it's submitted. If you need more, cancel and resubmit with a longer limit.

Need a starting point? TACC publishes copy-paste sample scripts per system: docs.tacc.utexas.edu/hpc/vista/#running

2

Monitor

Watch your job move through the queue, and read what its status is telling you.

2 Check the queue

Two tools show what your job is doing. `squeue` is the standard SLURM command; `showq` is a TACC utility many users prefer.

`squeue` — SLURM

```
squeue --me           # just my jobs
squeue -u netid       # a user's jobs
squeue --start        # est. start
-j 1234567            # one job
```

- ST column = job status code
- NODLIST/REASON shows why a job waits

`showq` — TACC

```
showq -u             # my jobs
showq                # everyone's
showq -U netid       # a user
showq -h              # help
```

- Groups jobs: ACTIVE, WAITING, BLOCKED,
- COMPLETING/ERRORED
- BLOCKED = can't run yet (e.g. maintenance)

2 Reading your job's status

The ST column in squeue uses short codes. The most common ones:

 Pending Waiting in the queue for resources	 Running Allocated a node and executing now	 Completing Finishing up and cleaning resources
 Completed Finished successfully (exit 0)	 Cancelled Ended by you or an administrator	 Failed Ended with a non-zero exit code

A job stuck in PD isn't broken — check the REASON column to see what it's waiting on.

2 Going deeper

Beyond the queue view, these help you inspect a live job, estimate a start, or review one that already finished.

```
scontrol show job 1234567
```

Full detail on one job: node list, time limits, why it's pending.

```
squeue --start -j 1234567
```

SLURM's best estimate of when a pending job will start.

```
sacct -j 1234567
```

Accounting record — works even after a job leaves the queue.

```
tail -f myjob.o1234567
```

Follow your output file live as the job writes to it.

3

Cancel

Pull a job back out of the queue — or kill a running one — with `scancel`.

3 Cancel a job with scancel

scancel works on both pending and running jobs. For a running job it sends a stop signal, waits briefly, then forces it to end.

```
scancel 1234567           # cancel one job by ID
scancel 1234567 1234568   # cancel several at once
scancel --name=myjob      # cancel by job name
scancel -u netid          # cancel ALL of your jobs
scancel --state=PENDING -u netid # only your queued jobs
```

Lost the job ID? Run `queue --me` or `showq -u` to look it up — the first column is the ID. Then scancel it.



TAP

The TACC Analysis Portal: launch interactive sessions from the browser — no scripts required.

TAP — the no-script path

The TACC Analysis Portal (tap.tacc.utexas.edu) launches interactive sessions — **Jupyter notebooks, RStudio, and DCV remote desktops** — without writing SBATCH. Any user with an allocation can log in with their TACC credentials and MFA.

-  **1 Log in** Go to tap.tacc.utexas.edu; sign in with your TACC username, password, and MFA.
-  **2 Choose System** Pick the machine where you hold an allocation (Vista, Frontera, ...).
-  **3 Pick Application** Jupyter, RStudio, or a DCV remote desktop.
-  **4 Set Job Details** Queue, node count, and run time — the same choices as #SBATCH.
-  **5 Submit** TAP queues the job; when it starts you get a link straight into the session.

+ TAP — monitor & end your session

While it runs

- The Jobs page shows each session's status and its connection link.
- A pending session behaves like a queued batch job — it waits for a node.
- Utilities › Run Sanity Tool catches common account issues (e.g. over quota on \$HOME).

When you're done

- End the session from TAP when finished — don't just close the tab.
- An open session keeps holding the node and charging your allocation until it ends or hits its time limit.
- Ending it is TAP's equivalent of scancel.

Same idea, either way: submit → monitor → end. TAP just gives you buttons instead of sbatch, squeue, and scancel.

Command cheat sheet

SUBMIT

```
scp -r folder user@vista.tacc.utexas.edu:
```

Copy files up to TACC

```
sbatch myjob.slurm
```

Submit a job; returns a job ID

MONITOR

```
squeue --me / showq -u
```

See your jobs in the queue

```
scontrol show job <id>
```

Full detail on one job

```
sacct -j <id>
```

History, even after it finishes

CANCEL

```
scancel <id>
```

Cancel one job by ID

No scripts? Launch Jupyter / RStudio / desktop sessions at tap.tacc.utexas.edu

```
scancel -u netid
```

Cancel all of your jobs